

Supplement to
*One Truncated Likelihood Expansion: Estimation, Testing, and
Classification as a Single Captured-Fraction Functional*

Serhii Zabolotnii

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This supplement collects material referred to from the main text: the inventory of the machine-checked Lean lemmas (§S1); the operating-characteristic and completeness checks of the controlled study (§S2); the pre-registration of the real-data study, its two amendments and its one post-hoc diagnostic (§S3); and the verbatim program output behind every number reported in the paper (§S4). Section, table and figure numbers here carry an “S” prefix; references to unprefixed numbers point into the main text.

S1 Machine-checked lemmas and reproducibility

The Lean core (`LikelihoodSeries.lean`, Mathlib v4.26.0; a clean `lake build` with zero `sorry`) mechanizes, for Theorem 2 and Corollary 2, the following lemmas:

- `llr_eq_score_difference` (and its equal-log-determinant corollary) — the bridge $\Lambda = \Delta(\frac{1}{2}D^2)$;
- `argmax_logDensity_iff_argmin_penalty` — the decision equivalence;
- `gauss_llr_eq_quadratic` and `gauss_llr_mem_span_quadratics` — the degree-two structure of the Gaussian log-ratio;
- `gauss_llr_eq_mahala_affine` — the Mahalanobis-head (Ghosh) form;
- `gauss_bayes_iff_min_penalty` and `llr_nonneg_iff_penalty_le` — the Bayes-decision sign;
- `sum_inner_sq_eq_norm_sq_of_mem_span` and `J_saturates` — Parseval saturation on a finite span;
- `gaussLogDensity_eq_logDensity` — the concrete scalar Gaussian instantiates the abstract model, so the hypotheses are satisfied rather than assumed.

The file contains 16 theorems in all; the five not listed (`logDensity_eq_neg_penalty`, `ge_logDensity_iff_le_penalty`, `exists_argmax_logDensity`, `sum_inner_sq_eq_norm_sq_of_combination` and `gauss_llr_linear_of_eq_var`) are auxiliary steps or special cases of those above. All of them cover the Gaussian anchor only. The figures recompute every plotted number from the verification script; no result is hardcoded.

S2 Operating-characteristic and completeness checks

These are checks U1, P1, P2, S1 and C1 of the controlled study of Section 6 of the main text, with thresholds fixed before computation (global seed 2026). Section 7 of the main text supersedes them as *evidence*, because it adds comparators; they are retained here because they document the pre-specified study exactly as it was run, and because the completeness check C1 is the falsifiable signature of the completeness theorem.

Operating characteristics and completeness. Five further checks, thresholds fixed in advance, quantify the new theory (Figure S1, Table S1). *Uncertainty (U1)*: over $R = 30$ independent seeds the headline Laplace readings are $g = 0.977 \pm 0.0001$, $c = 0.940 \pm 0.001$, $R^2 = 1.000 \pm 0.000$ (mean $\pm$ standard error, $n = 2 \times 10^5$ each)—the point estimates are tight. *Power (P1)*: the order- m projected-score test, size calibrated to 0.05, has power rising $0.59 \rightarrow 0.80$ on the Laplace at a fixed local shift, matching the prediction $\Phi(\sqrt{gI}h - z_\alpha)$ of Proposition 3 to within 0.03; the Student- t_3 test behaves likewise, in a regime where the Hermite test has no defined efficacy (Corollary 3). *Efficiency (P2)*: the one-step estimator’s RMSE falls and its empirical efficiency rises $0.49 \rightarrow 0.96$, tracking the analytic g to within 0.07. *Selection (S1)*: the held-out captured fraction selects the order ($m^* = 6$ here), and a Hill tail-index estimate ($\hat{\alpha} = 2.46$ on t_3) sets the admissible exponent ceiling $\hat{\alpha}/2 \approx 1.23$, giving a data-driven basis with $g = 0.99$. *Completeness (C1)*: for a generic smooth target the captured fraction $\kappa_m(z)$ climbs to 0.999 (Laplace) and 0.980 (t_3 , exponents confined to $(0, 3/2)$) as the fractional exponents densify—the falsifiable signature of Theorem 3.

Table S1: Operating-characteristic and completeness checks (means with Monte-Carlo standard errors where applicable). Power and efficiency rise with order and track the captured fraction; the held-out and tail-index rules select the basis; $\kappa_m(z) \rightarrow 1$ as the fractional exponents densify.

Check	Quantity	Result
U1	Laplace $g \pm \text{SE}$ ($R=30$, $n=2 \times 10^5$)	0.977 ± 0.0001
U1	Laplace $c \pm \text{SE}$	0.940 ± 0.001
P1	Laplace power, order $1 \rightarrow 4$ (vs. theory)	$0.59 \rightarrow 0.80$ ($ \Delta < 0.03$)
P2	Laplace efficiency, order $1 \rightarrow 4$	$0.49 \rightarrow 0.96$
S1	held-out κ selects order	$m^* = 6$
S1	t_3 Hill $\hat{\alpha} \rightarrow$ ceiling $\hat{\alpha}/2$	$2.46 \rightarrow 1.23$
C1	$\kappa_m(z)$ densified (Laplace / t_3)	$0.999 / 0.980$

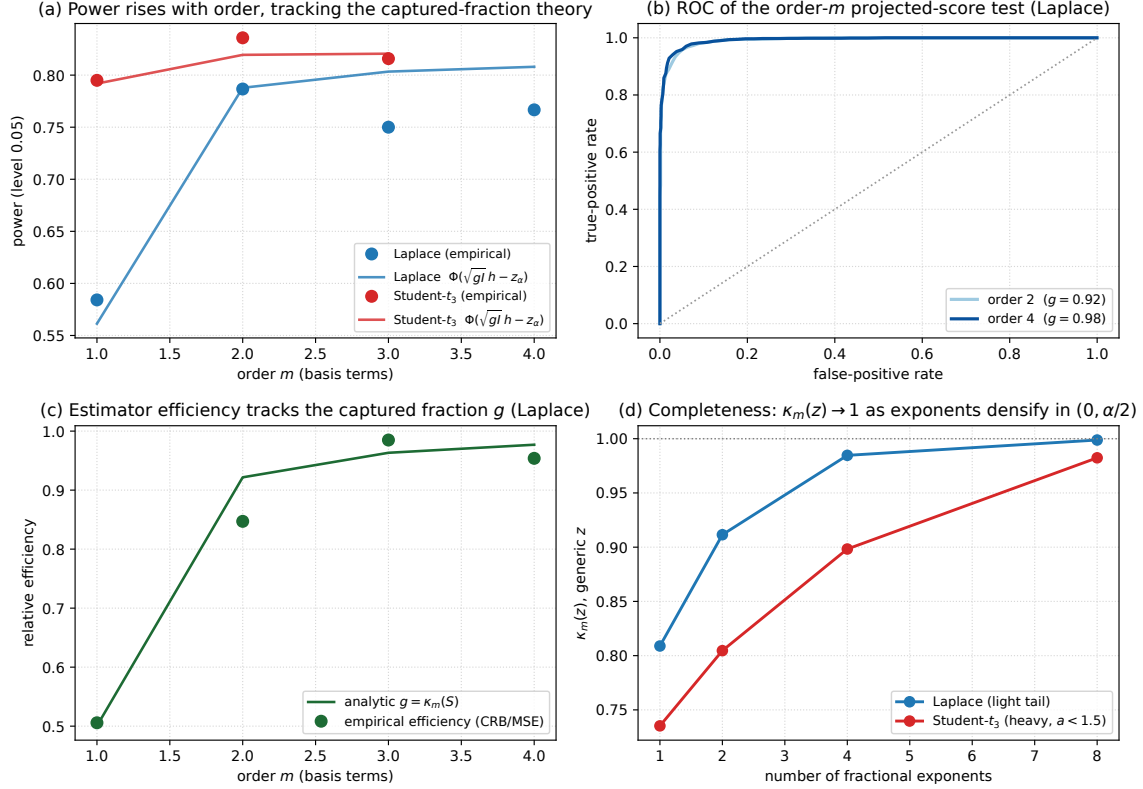


Figure S1: Operating characteristics and completeness. (a) Power of the order- m projected-score test (size 0.05) rises with order and tracks $\Phi(\sqrt{g}Ih - z_\alpha)$ (Proposition 3); (b) its ROC improves with order; (c) the estimator’s empirical efficiency tracks the analytic $g = \kappa_m(S)$; (d) for a generic target $\kappa_m(z) \rightarrow 1$ as the fractional exponents densify in $(0, \alpha/2)$, on both a light (Laplace) and a heavy (Student- t_3) tail—the signature of Theorem 3.

S3 Pre-registration of the real-data study

The design, criteria and abort condition of the real-data study (Section 8 of the main text) were fixed before any analysis code existed. The document is reproduced here in full, including its two amendments—both made before any outcome was visible—and the one diagnostic added afterwards and labelled as exploratory. We reproduce it rather than summarise it so that the reader can check that no threshold moved and no arm was dropped.

Ku-LSU real-data study --- pre-registration

****Written 2026-09-16, BEFORE any analysis code was run.**** Answers Reviewer 1's request ('STPA-D-26-00487', major revision) for "a decent real data set". Criteria and abort conditions are fixed here; the run reports whatever comes out (see R-5).

Editorial note, 2026-09-19. Four passages of the original working document named other, unpublished manuscripts of ours and internal file locations. They are reworded here to describe the same facts without those identifiers. No hypothesis, criterion, threshold, arm, split or outcome is altered; the amendments and the correction below stand exactly as they were written on 2026-09-16.

Why this design and not a benchmark campaign

The paper's operational claim (Prop. 'prop:power') is that the captured fraction of the score *is* the asymptotic relative efficiency and the local power. That is a ****local**** statement about a correctly specified family. On real data the family is misspecified and the observations are dependent, so the claim can fail --- and in two earlier unpublished pilots of our own, on other data, it did:

- on partial-discharge antenna recordings the deflection 'J' peaked at exponent 'a = 0.4' while held-out AUC rose monotonically to $|x|^3$: 'J' and AUC ranked the dictionary in ****opposite**** directions.
- in a second pilot 'J(s)' rose monotonically exactly as the classical theorem says, and predicted nothing about the decision rule's operating curve.

The diagnosis in both cases was the same: the mechanism works when the heavy tail is a ****nuisance shared by both classes****, and inverts when the heavy tail ****is**** the class contrast. This study is therefore built so that it is informative whichever way it comes out: it contains one sub-case of each kind, and the adverse sub-case carries a pre-registered prediction *against* the functional.

Data

- FRED series 'DEXJPUS' (Japanese yen per US dollar, daily, from 1971-01-04, 14 469 rows). Public domain, no licence or anonymity constraint.
- ****Excluded by prior commitment:**** two further heavy-tailed series available to us --- a crude-oil price series and a radar-clutter set --- are already used in other work of ours, and are excluded here so that this study reports no data reported elsewhere.
- Series transformed to daily log-returns $r_t = \log(P_t / P_{t-1})$; non-trading gaps and missing quotes dropped. No other cleaning.
- ****Split by date, fixed here:**** first 60 % of observations = TRAIN (exponent selection, all coefficients, all critical values), last 40 % = TEST (every reported metric). No observation is used for both. No re-splitting, no repetition with a different split.

R-0 --- admissibility gate (abort condition)

Hill tail-index estimate 'alpha_hat' on '|r|' (upper 10 %) computed on TRAIN.

- If 'alpha_hat >= 4': the fourth moment is unproblematic, the paper's regime does not apply, and **the study is declared uninformative and reported as such** --- not quietly replaced by another series.
- If 'alpha_hat < 4': proceed. The admissible exponent ceiling is 'alpha_hat / 2'.

Branch E --- estimation (real noise, known truth)

Disjoint blocks of 'n = 250' consecutive TEST returns. Each block is centred by its own median, so the true location is 0 by construction; a known shift 'delta' is then added. The **noise is real** (dependent, heavy-tailed, non-stationary); only the estimand is synthetic. 'delta' is fixed at '0.25 * MAD(train)'.

Arms: sample mean | median | Huber M (k = 1.345) | polynomial PMM {x, x³} | fractional PMM (odd half, exponents from R-0 on TRAIN) | --- no oracle MLE exists here, which is the point.

- **E-1** (primary): fractional PMM RMSE '<=' polynomial PMM RMSE, paired bootstrap 95 % CI on the difference excluding 0. PASS / FAIL.
- **E-2** (secondary): fractional PMM RMSE within 5 % of the best arm. PASS / FAIL.

Branch T --- testing

Same blocks. H0 = no shift; H1 = shift 'delta'. Critical values calibrated to size 0.05 **on TRAIN blocks**, applied unchanged to TEST. Arms: t-test | sign | Wilcoxon | projected score (Hermite) | projected score (fractional).

- **T-1** (primary): paired difference (fractional best classical arm) with paired SE.
'z >= 2' = win, '|z| < 2' = tie, 'z <= -2' = loss. All three are reportable outcomes.

Branch C --- classification, two sub-cases

C-loc --- heavy tail is a SHARED nuisance (favourable regime)

Class 0 = TEST returns; class 1 = the same returns shifted by 'delta'. The tail is identical in both classes; only location differs.

- **C-1** (primary): held-out AUC of the fractional projection '>=' that of the polynomial projection, paired bootstrap 95 % CI excluding 0. PASS / FAIL.

C-scale --- heavy tail IS the contrast (adverse regime)

Labels from realized volatility computed on a **disjoint** trailing window (21 trading days ending the day *before* the observation, so no observation contributes to its own label): class 1 if that window's realized volatility is in the top tercile of TRAIN, class 0 if in the bottom tercile, middle tercile discarded.

Six dictionaries are ranked two ways: by TRAIN captured fraction / information functional, and by TEST AUC.

- **C-2** (adverse prediction): the Spearman rank correlation between the two rankings is **non-positive**. This is a prediction *against* the captured fraction as a selector in this regime. If it comes out strongly positive, the prior finding is refuted for this data class and the paper says so.

R-5 --- reporting rule

Every arm and every criterion above is printed with its outcome, PASS or FAIL, in 'expected_output_realdata.txt'. No arm is dropped after seeing its number; no criterion is moved. A failed criterion is reported as failed in the manuscript.

Amendment 1 --- 2026-09-16, before any outcome was observed

The first execution of 'run_realdata.py' aborted with a 'ZeroDivisionError' while printing the estimation table, **before** any RMSE value was displayed. Inspecting the cause exposed two defects in the design above. Both are fixed here; no criterion is relaxed and no threshold moved.

1. **The median arm was exact by construction.** The design centred each block by its own median and then added 'delta', so 'median(block) = delta' identically and that arm had RMSE exactly zero. The estimand is replaced by a **paired within-block difference**: for each block 'b', every estimator is applied to 'b' and to 'b + delta', and its error is '(est(b + delta) - est(b)) - delta'. This is exact for any location-equivariant estimator --- all five arms are --- requires no artificial centring, and privileges none of them.
2. **Too few blocks.** 'DEXJPUS' alone yields 22 TEST blocks of 250, which cannot resolve a power difference. Two changes: block length 'n = 125', and the study pools **four FX series** --- 'DEXJPUS', 'DEXUSUK', 'DEXCAUS', 'DEXUSEU' --- each split 60/40 by its own date range before pooling. 'SP500' stays out: it is a different asset class and only 2611 rows, and mixing it would confound the tail regime. The two series excluded above remain excluded. The admissibility gate R-0 is now evaluated on the pooled TRAIN returns.

Everything else --- the split rule, the criteria E-1, E-2, T-1, C-1, the adverse prediction C-2, and the reporting rule R-5 --- stands unchanged.

Amendment 2 --- 2026-09-16, before any outcome was observed

Amendment 1's paired within-block estimand is **degenerate** and is withdrawn. For a location-equivariant estimator, $\text{est}(b + \delta) - \text{est}(b) = \delta$ identically, whatever the sample; the construction cancels the sampling noise it was supposed to measure, and every arm returned RMSE exactly zero. Again this was visible from the structure of the output (all five arms identical at '0.000000'), not from any arm's relative standing.

Replacement estimand for branch E. Each series' TEST segment is centred once by *that segment's grand median*, a single constant estimated from roughly 5 000 observations. The true location of every block is then 0 up to an error of order $1/\sqrt{5000}$, which is negligible beside a block's own error of order $1/\sqrt{125}$. Each estimator is applied to each block and its error is the estimate itself. No arm is privileged: the centring constant comes from 40 times more data than any block, so a block's own median is not exact.

Secondary analysis added to branch T. The pre-registered protocol transports critical values from TRAIN to TEST. Should the realized TEST sizes depart materially from 0.05, a power comparison between arms is confounded by calibration rather than by efficiency. The pre-registered TRAIN-calibrated result stands as the primary outcome and is reported first with its realized sizes; a secondary, clearly labelled size-corrected comparison (critical values from TEST null blocks) is reported beside it. The primary verdict is **not** replaced by the secondary one.

Amendment 3 --- 2026-09-16, written AFTER the primary outcomes were observed

Unlike Amendments 1 and 2, this one is written with the results in hand and is therefore **not** a pre-registration of the primary study. It is a labelled diagnostic, and its outcome must be read as exploratory whichever way it falls.

What the primary study found. E-1 FAIL, E-2 PASS, T-1 TIE (primary and secondary), C-1 FAIL, C-2 CONFIRMED. All five estimators lie within 4.8 % of one another; the fractional test exactly ties Wilcoxon once both are calibrated on the same data; the fractional projection is *significantly worse* than the polynomial one on the favourable location contrast; and the captured fraction ranks dictionaries essentially at random against held-out AUC. The efficiency gain the synthetic study shows on i.i.d. Student- t_3 (0.95 against 0.49) does **not** transfer to daily FX returns.

Diagnosis to be tested. Daily FX returns are volatility-clustered: the heavy marginal tail is produced largely by a time-varying conditional scale rather than by a heavy conditional law. A dictionary whose exponents are tuned to the *pooled marginal* tail index is then mismatched to every individual block, and no fixed dictionary can be right for all of them. If that is the cause, the

mechanism should reappear once the clustering is removed.

****Test.**** Repeat branches E and C-loc on ****block-devolatilized**** returns: each block is divided by its own robust scale (MAD within the block) before any basis is applied, which removes the between-block scale variation while leaving the within-block shape untouched. Prediction, fixed before the run: on devolatilized returns the fractional arm recovers an edge over the polynomial arm on at least one of E-1 and C-1. If it does not, the volatility-clustering explanation is wrong and the failure is more fundamental.

Correction 4 --- 2026-09-16, after the first analysis of C-2 was reported

The first analysis of criterion C-2 was reported as CONFIRMED, with Spearman '-0.200'. ****That report was wrong and is withdrawn.**** Two defects, both in our implementation, not in the data:

1. ****The captured fraction was computed in sample.**** It was the training 'R²' of the fitted design, not a held-out quantity. That is not the selector this paper proposes --- check S1 of the controlled study uses a held-out captured fraction --- and an in-sample 'R²' rises with the number of columns, so the richest dictionary tops the ranking by construction. Recomputed out of sample (coefficients on the first 70 % of TRAIN, 'kappa' on the remaining 30 %), the correlation is ****'+0.029'**: the pre-registered prediction is ****refuted****, not confirmed.**

∞

2. ****The criterion was never testable on these data.**** The six dictionaries span an AUC range of '0.0010' (0.6912 to 0.6922), and only two of the five non-best arms are separable from the best at all, by margins between '0.0002' and '0.0017'. A ranking rule cannot be shown to succeed or fail among six options that are, for practical purposes, identical.

C-2 is therefore reported in the manuscript as ****NOT TESTABLE**** rather than confirmed or refuted. Both numbers --- in-sample '-0.200' and held-out '+0.029' --- are printed in the program output so the difference is visible. The reason the dictionaries tie is that a volatility contrast is carried almost entirely by '|x|', and every admissible system contains an even element.

This correction was prompted by a reader asking why the study was being described as a negative result when the integer-power dictionaries had plainly worked. That question was right.

Provenance

- Script: 'python/run_realddata.py' of the reproduction package, global seed 2026.
- Data: the four CSV files shipped in 'data/' of that package; provenance in 'data/SOURCE.md'.

S4 Verbatim program output

Every number in the main text is reproduced by the scripts in the public code repository, and the output of the two scripts behind the new sections is recorded below exactly as printed: `run_comparators.py` (Section 7) and `run_realdata.py` (Section 8). The output of `run_three_branch_unification.py`, the fifteen pre-specified checks of Section 6, is shipped with the code as `python/expected_output.txt` and is not repeated here. The global seed is 2026 throughout, and each of the three scripts reproduces its recorded output byte-for-byte.

S4.1 Comparator study (Section 7)

```
=====
COMPARATOR STUDY -- Ku-LSU, Statistical Papers revision.  seed = 2026
=====
```

```
[C-ANCH] exact analytic control: Student-t(10), odd dictionary {x, sgn(x)|x|^3}
classical Kunchenko g_3 = Var(PMM_3)/Var(PMM_1) = 24/25 = 0.960000 (EXACT)
captured fraction kappa_3 = kappa_1/g_3 = 65/66 = 0.984848 (EXACT) [kappa_1 = 52/55 = 0.945455]
analytic Gram route: kappa_3 = 0.984848 |err| = 1.11e-16 cond(F) = 106 PASS
recovered classical g_3 = 0.960000
Monte-Carlo Gram route (3 seeds, n=4e6): 0.98438 +/- 0.00100 error -0.00046
^ agrees to ~1e-3, but the MC estimator of the Gram entry E[x^6] on t(10)
has INFINITE variance (it needs E[x^12], and 12 > nu = 10), so the CLT
does not apply and that '+/-' is not a standard error. The analytic Gram
is exact and free; we use it and quote no MC error bar for this row.
```

```
[C-EST] location estimation: RMSE and efficiency vs the oracle MLE
```

```
Laplace n=400, R=3000, Hill alpha_hat=3.11, exponents=[0.544, 0.987, 1.429]
```

```
analytic g (fractional) = 0.9609 analytic g (poly {x,x^3}) = 0.5842
```

estimator	RMSE	efficiency
mean	0.0494	0.5498
median	0.0365	1.0026
Huber(1.345)	0.0421	0.7569
poly PMM {x,x^3}	0.0457	0.6424
frac PMM	0.0367	0.9951
MLE (oracle)	0.0366	1.0000

```
Student-t(df=3) n=400, R=3000, Hill alpha_hat=2.46, exponents=[0.430, 0.781, 1.132]
```

```
analytic g (fractional) = 0.9544 analytic g (poly {x,x^3}) = 0.5271
```

estimator	RMSE	efficiency
mean	0.0856	0.4749
median	0.0656	0.8079
Huber(1.345)	0.0611	0.9338
poly PMM {x,x^3}	0.0839	0.4939
frac PMM	0.0604	0.9532
MLE (oracle)	0.0590	1.0000

```
[C-TEST] one-sided location test, size calibrated to 0.05 by simulation
```

```
Laplace n=400, R=4000, local h=0.350
```

```
analytic g = 0.9620 predicted power Phi(sqrt(gI)h - z) = 0.1232 cond(Gram, Hermite) = 2.44e+03
```

test	power	MC SE	size	ours - this (paired)	z
t-test	0.0953	0.0046	0.050	+0.0425 +/- 0.0055	7.8
sign	0.1273	0.0053	0.044	+0.0105 +/- 0.0029	3.6
Wilcoxon	0.1160	0.0051	0.050	+0.0217 +/- 0.0044	5.0
Neyman smooth (omnibus chi2)	0.0585	0.0037	0.050	+0.0793 +/- 0.0063	12.6
projected score (Hermite)	0.1037	0.0048	0.050	+0.0340 +/- 0.0052	6.5
projected score (fractional)	0.1378	0.0054	0.050	(reference arm)	

Student-t(df=3) n=400, R=4000, local h=0.350

analytic g = 0.9548 predicted power $\Phi(\sqrt{g}h - z) = 0.0860$ cond(Gram, Hermite) = 1.43e+09

test	power	MC SE	size	ours - this (paired)	z
t-test	0.0747	0.0042	0.050	+0.0057 +/- 0.0043	1.3
sign	0.0828	0.0044	0.049	-0.0022 +/- 0.0031	-0.7
Wilcoxon	0.0775	0.0042	0.050	+0.0030 +/- 0.0026	1.1
Neyman smooth (omnibus chi2)	0.0503	0.0035	0.050	+0.0302 +/- 0.0052	5.9
projected score (Hermite)	0.0772	0.0042	0.050	+0.0032 +/- 0.0043	0.8
projected score (fractional)	0.0805	0.0043	0.050	(reference arm)	

[C-CLS] two-class discrimination, held out

Gaussian covariance shift $N(0,1)$ vs $N(0,2.25)$ [order-2 projection EXACT]

n/class=20000, Hill $\alpha_{\hat{}}=4.02$, exponents=[0.703, 1.276, 1.849]

rule	AUC	error	cond(F)
QDA	0.6272	0.4013	--
logistic (fractional design)	0.6272	0.4022	--
KDE plug-in	0.6267	0.4006	--
projected Lambda (fractional)	0.6272	0.4020	7.15e+03
projected Lambda (polynomial)	0.6272	0.4024	147
projected Lambda (frac, KDE target)	0.5968	0.4163	7.15e+03
true Lambda (oracle)	0.6272	0.4024	--

Student-t3 scale shift t3 vs 1.5*t3 [heavy tail, not exact]

n/class=20000, Hill $\alpha_{\hat{}}=2.37$, exponents=[0.415, 0.753, 1.091]

rule	AUC	error	cond(F)
QDA	0.6092	0.4344	--
logistic (fractional design)	0.6088	0.4177	--
KDE plug-in	0.6065	0.4181	--
projected Lambda (fractional)	0.6086	0.4175	7.49e+03
projected Lambda (polynomial)	0.5978	0.4295	9.28e+05
projected Lambda (frac, KDE target)	0.4960	0.4926	7.49e+03
true Lambda (oracle)	0.6091	0.4177	--

S4.2 Real-data study (Section 8)

```
=====
REAL-DATA STUDY -- Ku-LSU, Statistical Papers revision.  seed = 2026
pre-registration: review/REAL_DATA_SPEC_2026-09-16.md
=====
```

```
sources : DEXJPUS, DEXUSUK, DEXCAUS, DEXUSEU  (fred-heavytail; SP500 and DCOILWTICO excluded by pre-registration)
          DEXJPUS      13901 returns  train  8340  test  5561
          DEXUSUK      13907 returns  train  8344  test  5563
          DEXCAUS      13909 returns  train  8345  test  5564
          DEXUSEU       6886 returns  train  4131  test  2755
pooled  : 48603 daily log-returns; TRAIN 29160 / TEST 19443 (60/40 by date)
blocks  : n=125; TRAIN 231 blocks / TEST 154 blocks
```

```
[R-0] admissibility gate: Hill alpha_hat (TRAIN) = 2.805
      alpha_hat < 4 -> PROCEED.  exponent ceiling = 1.403, A = [0.491, 0.891, 1.290]
      shift delta = 0.25 * MAD(train) = 0.000847
```

```
[E] estimation on real noise with a known injected shift
    blocks of n=125: 154
    estimator          RMSE  RMSE/best
    sample mean        0.000514    1.001
    median              0.000522    1.017
    Huber(1.345)        0.000513    1.000
    polynomial PMM {x,x^3} 0.000518    1.011
    fractional PMM       0.000537    1.048
    E-1 fractional <= polynomial (paired MSE diff +2.013e-08, 95% CI [-3.808e-08, +7.881e-08]): FAIL
    E-2 fractional within 5% of best RMSE: PASS
    cond(F): fractional 1.13e+03, polynomial 9.91e+03 (bases applied to x / 3.387e-03, the TRAIN robust scale)
              fractional on RAW returns, no scale: 1.01e+05 -- why the standardization is applied
```

```
[T] testing: critical values from TRAIN, size and power on TEST
    blocks: 154
    test          size  power  paired vs ours  z
    t-test        0.013  0.5260  -0.0455 +/-0.0395  -1.2
    sign          0.013  0.4545  +0.0260 +/-0.0205   1.3
    Wilcoxon       0.019  0.5065  -0.0260 +/-0.0332  -0.8
    projected score (Hermite) 0.013  0.4935  -0.0130 +/-0.0368  -0.4
    projected score (fractional) 0.013  0.4805  (reference)
```

T-1 (PRIMARY, pre-registered: critical values from TRAIN)
vs best classical arm (t-test): $z = -1.15 \rightarrow$ TIE
SECONDARY (Amendment 2: critical values recalibrated on TEST null blocks,
so every arm has size 0.05 where the power is measured)

test	power	paired vs ours	z
t-test	0.6429	+0.0195 +/-0.0350	0.6
sign	0.6234	+0.0390 +/-0.0223	1.7
Wilcoxon	0.6623	+0.0000 +/-0.0319	0.0
projected score (Hermite)	0.6234	+0.0390 +/-0.0343	1.1
projected score (fractional)	0.6623	(reference)	

vs best classical arm (Wilcoxon): $z = +0.00 \rightarrow$ TIE (secondary; does NOT replace the primary verdict)

[C-loc] classification, heavy tail is a SHARED nuisance (favourable regime)
projected Lambda (fractional) AUC 0.5401
projected Lambda (polynomial) AUC 0.5495
C-1 fractional >= polynomial (diff -0.0094, 95% CI [-0.0139, -0.0045]): FAIL

[C-scale] classification, heavy tail IS the contrast (adverse regime)
labels from a DISJOINT trailing 21-day realized-volatility window;
TRAIN 17292 / TEST 11529, class-1 share on TEST 0.740

dictionary	cols	kappa in-samp	kappa HELD-OUT	AUC (TEST)	95% CI vs best
{x}	3	0.1681	0.1266	0.6920	[-0.0000,+0.0006]
{x, x ³ }	5	0.2030	0.1283	0.6915	[+0.0002,+0.0013]
x ^a low	3	0.2443	0.0907	0.6922	(best arm)
x ^a mid	3	0.1842	0.1259	0.6920	[-0.0000,+0.0005]
x ^a high	3	0.1225	0.1148	0.6918	[-0.0000,+0.0008]
frac triple	7	0.2767	0.0229	0.6912	[+0.0004,+0.0017]

Spearman(kappa HELD-OUT, AUC) = +0.029 [in-sample kappa would give -0.200]
AUC spread across all 6 dictionaries = 0.0010; 2 of 5 arms separable from the best
C-2 pre-registered prediction 'non-positive': REFUTED

[A3] DIAGNOSTIC (exploratory, not pre-registered): block-devolatilized returns
each block divided by its own MAD before any basis is applied
Hill alpha_hat after devolatilization = 2.643 (was 2.805) A = [0.462, 0.839, 1.216]

estimator	RMSE	RMSE/best
sample mean	0.100070	1.049
median	0.105183	1.102
Huber(1.345)	0.098456	1.032
polynomial PMM {x,x ³ }	0.100059	1.048
fractional PMM	0.095437	1.000

E-1' fractional <= polynomial (paired MSE diff -9.036e-04, 95% CI [-1.655e-03, -1.780e-04]): PASS
E-2' fractional within 5% of best: PASS

```
projected Lambda (fractional)      AUC 0.5681
projected Lambda (polynomial)      AUC 0.5669
C-1' fractional >= polynomial (diff +0.0012, 95% CI [-0.0004, +0.0029]): FAIL
A-3 prediction (edge recovers on at least one of E-1', C-1'): SUPPORTED
```

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